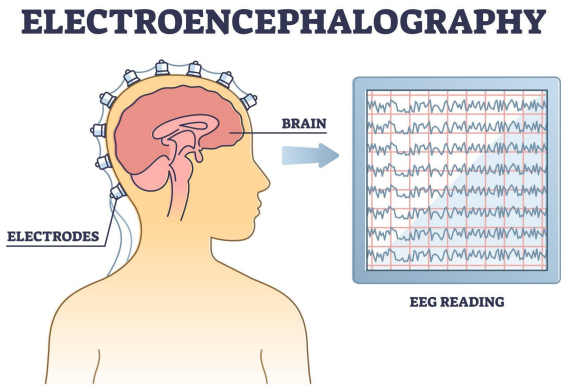
EEG-Based Machine Learning for Epileptic Seizure (Team 3)

By Holland, Kushal, Stephen, Donovan

The Problem

- Epileptic seizures are a result of sudden abnormal electrical activity in the brain.
- Doctors have been able to use a technique called EEG to measure electrical activity in people's brains
- **Problem**: Manual review of EEG recordings is slow, difficult, and labor-intensive.
- **Goal:** Automatically detect seizure events from multichannel EEG data using machine learning.



Why This Matters

- Faster and more reliable seizure detection can improve care
- Automated systems reduce burden on neurologists and EEG technicians
- Early detection can help improve:
    - Real-time monitoring in hospitals
    - At-home seizure alert systems
    - Long-term patient tracking
- Machine learning provides a scalable way to identify neural patterns not visible to humans

Approach & Methodology

Overview of Our Pipeline

Five major components:

1. Data loading (EDF → raw signals)
2. Preprocessing & filtering
3. Window segmentation & labeling
4. Feature engineering (classical & deep learning)
5. Baseline and CNN model development

This pipeline transforms multi-hour EEG recordings into model-ready training samples.

Dataset Used: CHB-MIT

- 23 pediatric epilepsy patients
- 198 recorded seizures
- 23 EEG channels sampled at 256 Hz
- Multi-hour continuous EEG sessions (EDF file)
- Precise annotation files provide seizure onset & offset times
- Widely used benchmark dataset for seizure detection research

Preprocessing Steps

- **Bandpass filtering (0.5–40 Hz):** removes drift + high-frequency noise
- **Artifact reduction:** muscle + environmental noise via FIR filtering
- **Normalization:** each window normalized per channel
- **Resampling to 256 Hz:** standardizes recordings
- Ensures consistent, noise-reduced signals for all models

Windowing & Labeling

- Split recordings into **4-second windows** with **50% overlap**
    - 4 sec × 256 Hz = 1024 samples per channel
- Output tensor shape: **(23 channels × 1024 samples)**
- Label = **1** if window overlaps any part of a seizure
- This results in **tens of thousands of labeled samples** per patient
- Windowing enables supervised learning and avoids information leakage

Feature Engineering Approaches

Classical ML Features

- Windows flattened into feature vectors

Deep Learning Features

- **1D CNN:** learns temporal patterns across channels
- **2D CNN:** learns time–frequency structure from spectrograms
- Best-performing architecture from early experiments

Modeling Overview

Baseline Models

- Logistic Regression
- Support Vector Machine (RBF)
- Random Forest (balanced class weights)
- Evaluate limitations of traditional ML on imbalanced EEG tasks

Deep Learning Models

- **1D CNN:** channel-temporal convolution
- **2D CNN:** spectrogram-based CNN for seizure pattern recognition
- Use patient-independent train/val/test split

Data & Experiments

Data Summary



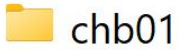
Database Open Access

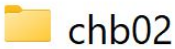
CHB-MIT Scalp EEG Database

- 24 pediatric epilepsy patients
- 198 recorded seizures
- Real clinical data
- High variability good for ML models

Train/Validation/Test Split

Patient-independent splitting

Train on one set of patients -> 

Test on entirely different patients -> 

Why use patient-independent splitting?

- Prevents information leakage
- More realistic evaluation for clinical use
- Much harder than random splitting → exposes generalization problems

Class Imbalance

- Seizure windows = ~3–6% of all windows
- Non-seizure windows dominate (normal EEG)
- Baseline models learn to always predict non-seizure
- Metrics like accuracy become misleading
- Sensitivity, specificity, and AUC are essential

```
Starting training...
--------------------------------------------------------------------
Epoch 1/50
  Train Loss: 0.0048, Train Acc: 0.9990
  Val Loss: 0.0015, Val Acc: 0.9999, Val AUC: 0.4897
  ✓ Saved best model (Val AUC: 0.4897)
```

Experiments Run

Baseline Model Experiments

- Logistic Regression
- SVM (RBF kernel)
- Random Forest

Deep Learning Experiments

- 1D CNN on raw EEG windows
- 2D CNN on spectrogram representations

Key Questions Tested

1. How do classical ML models handle EEG complexity?
2. Do CNNs learn meaningful seizure features?
3. Can models generalize across unseen patients?

Results & Findings

Baseline Model Results

Model	Accuracy	Sensitivity	Specificity	AUC
Random Forest	0.999	0.000	1.000	0.615
SVM	0.999	0.000	1.000	0.567
Logistic Regression	0.996	0.000	1.000	0.506

Observation:

- All baseline models show **high accuracy (~0.99)**
- But **sensitivity = 0.000** → they detect *zero seizures*
- This means they are predicting only the majority class

Conclusion:
Classical ML ≠ sufficient for seizure detection under imbalanced, patient-independent conditions.

Why Baselines Fail

Seizures are rare → majority class dominates

EEG patterns are nonlinear + high-dimensional

Baseline features cannot capture:

- temporal patterns
- spatial channel interactions
- frequency structure

Result: baselines overfit to non-seizure EEG

CNN Models Perform Better

Findings:

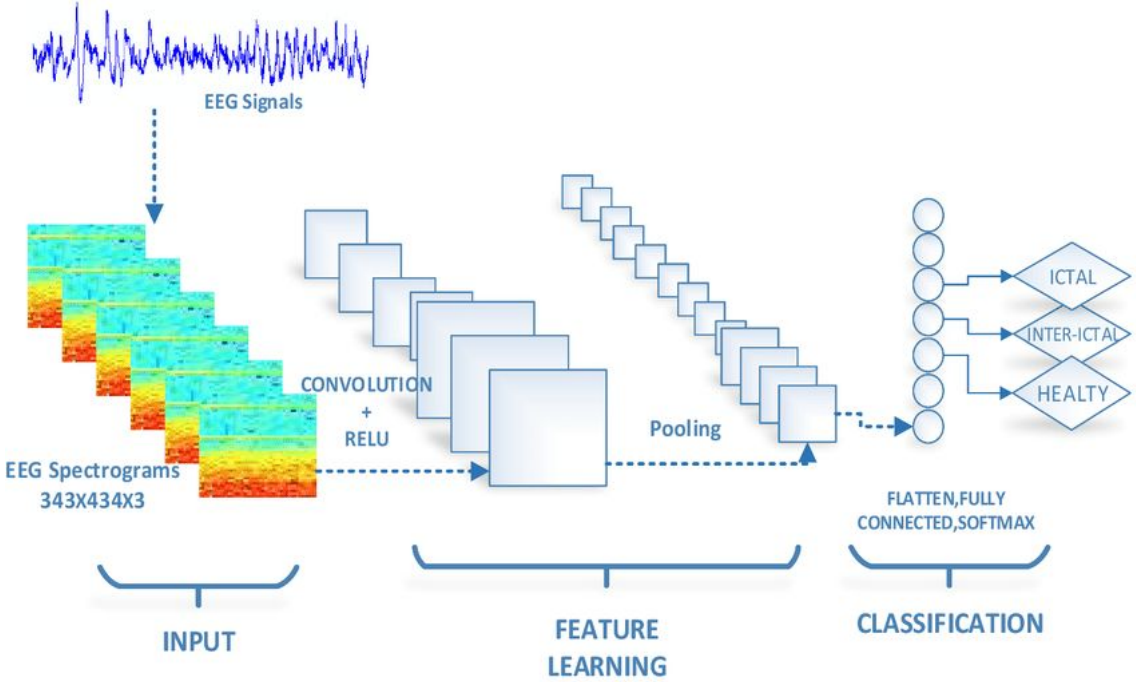
- CNNs learn discriminative features from EEG signals
- 1D CNN captures temporal patterns across channels
- 2D CNN learns time–frequency structure from spectrograms

Early 2D CNN performance:

- Much higher sensitivity vs. baselines
- Detects seizure windows that classical ML misses
- Still challenged by cross-patient generalization

Example: Spectrogram Input

CNN uses these time–frequency patterns to identify seizure onsets



Key Insights and Findings

Key Insights

- EEG seizure detection is a high-imbalance, high-variability problem, which makes classical ML models predict only non-seizure events
    - Accuracy is misleading
    - It is much more important that it can classify seizure events
- CNNs perform better because they can learn complex temporal and spatial patterns
- Cross-patient generalization remains the biggest challenge
- Diverse training data and strong preprocessing are essential for improving model reliability and real-world applicability

Limitations

- Limited number of patients in training data
  - 23 patients
- High patient variability
  - Seizure patterns differ widely across individuals
- Restricted experimentation
  - Deeper Models
  - Longer Training runs

Future Work

- Train CNNs on more patients
- Explore hybrid architectures such as CNN + LSTM or transformer-based models
- Introduce focal loss or class reweighting
  - Pay more attention to events labeled seizures
- Move toward a real-time seizure detection pipeline